

Diagnosing Cultural Bias in AVA-Trained Aesthetic Scoring: A Ground-Truth-Free Framework

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Abstract—Deep-learning models that score the aesthetic quality of photographs are typically trained on Western-centric datasets such as AVA. This raises a natural question: do these models behave differently on photographs from non-Western cultures? Answering it directly is hard, because non-Western photographs do not come with ground-truth aesthetic ratings, so accuracy cannot be measured. We sidestep this by measuring model behaviour instead. A ResNet18 model is fine-tuned on AVA, frozen, and run on curated photographic evaluation sets from three non-Western regions: East Asia, South Asia, and the Middle East and North Africa (MENA). All evaluation sets contain photography, like AVA, so that any differences can be attributed to cultural content rather than artistic medium. We examine the frozen model along three axes: shifts in predicted score distributions, per-image prediction confidence, and spatial attention measured via Grad-CAM using a border-to-centre activation ratio and attention heatmap entropy. Predicted-score distributions on the non-Western sets are highly significantly shifted and compressed relative to AVA; the model’s confidence is essentially unchanged; and Grad-CAM attention is systematically more diffuse on non-Western images. The model produces shifted scores and diffuse attention without flagging any increase in uncertainty, a combination with clear implications for the deployment of AVA-trained aesthetic models.

Keywords—Cultural bias, aesthetic assessment, ResNet18, domain shift, Grad-CAM, cross-cultural generalization, AVA dataset, computer vision ethics

I. INTRODUCTION

Automatic aesthetic assessment, the task of having a computer predict how visually appealing a photograph is to people, has become a common component of many digital systems. Models that score aesthetic quality are used in image search, photo organization, recommender systems, content moderation, and creative tools. The standard recipe is supervised learning: collect a large dataset of photographs paired with human ratings, train a convolutional neural network on them, and use the trained model to score new images. Benchmark performance on this task has improved steadily over the years.

This progress relies, however, on a small number of training datasets, of which the Aesthetic Visual Analysis (AVA) dataset [1] is the most widely used. AVA was collected from a single online photography community, dpchallenge.com, whose contributors and raters are predominantly Western. The dataset therefore encodes a culturally specific notion of what makes a “good” photograph, including conventions of composition, color, subject matter, and framing. A model trained

on AVA inherits these conventions implicitly. As we discuss in Section II, dataset bias research suggests such models may generalize poorly to under-represented populations, yet whether AVA-trained aesthetic models exhibit this has received little attention.

Testing this empirically is harder than it sounds. The most natural framing, “measure the model’s accuracy on non-Western images,” is not actually available to us: those images do not have ground-truth aesthetic ratings, and collecting them at the scale and reliability of AVA’s per-image rating distributions is not feasible for a focused study. We argue that accuracy is not the only useful signal of bias. A model that has internalized a narrow aesthetic distribution can be diagnosed by looking at its *behaviour* on out-of-distribution inputs: how its predicted-score distribution shifts, how its confidence changes, and where in the image it looks when making a judgment. All three can be measured directly from the model and require no ratings on the evaluation data.

In this paper, we develop and apply such a diagnostic framework. We fine-tune a ResNet18 aesthetic regressor [2] following the NIMA formulation [3] on AVA, freeze the resulting checkpoint, and treat it as a fixed measuring instrument. We then build openly-licensed photographic evaluation sets from three non-Western regions: East Asia, South Asia, and the Middle East and North Africa. The frozen model is examined along three axes: predicted score distribution shift, prediction confidence, and spatial attention via Grad-CAM [4]. Full details of each metric are given in Section IV. Our contributions are: (1) a ground-truth-free diagnostic framework for cultural bias in aesthetic models; (2) three curated, openly-licensed non-Western photographic evaluation sets, released with full provenance for reproducibility; (3) empirical findings on the AVA-trained ResNet18 baseline along the three diagnostic axes; and (4) a reproducible benchmark intended to support future work on more inclusive aesthetic models.

II. RELATED WORK

A. Aesthetic Assessment with Deep Networks

Deep learning approaches to image aesthetic assessment have progressed from simple binary classification to predicting full rating distributions. Early models used pre-trained networks like AlexNet and VGG to classify images as “high” or “low” quality [5]–[7], before later work shifted to predicting

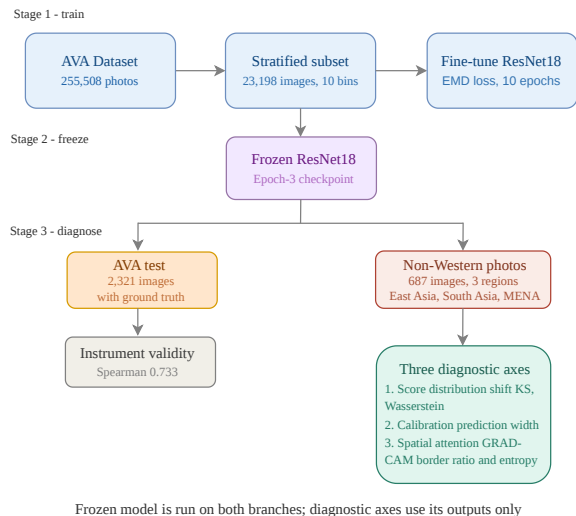


Fig. 1. Diagnostic pipeline. A ResNet18 aesthetic regressor is fine-tuned on a stratified subset of AVA and frozen at the checkpoint with lowest validation loss. The frozen model is then run on (a) the held-out AVA test set, to verify instrument validity, and (b) 687 curated non-Western photographs across three regions, to diagnose the model’s behaviour along three axes: predicted-score distribution shift, calibration, and spatial attention.

continuous aesthetic scores [8]. A key advancement came with the NIMA model [3], which predicts a full probability distribution over scores from 1 to 10 using Earth Mover’s Distance (EMD) loss. This approach better captures the ordinal nature of aesthetic ratings and reflects prediction uncertainty. Further extensions introduced multi-task architectures combining aesthetic scoring with attribute prediction [9] and attention mechanisms for capturing spatial interactions [10]. Despite these advances, and as surveyed by Daryanavard Chounchenani et al. [11], most models are trained on Western-centric datasets, leaving cross-cultural generalization largely unexplored.

B. Dataset Bias and Geographic Generalization

Aesthetic preferences vary systematically across cultures [11], yet most models are trained on Western-centric datasets such as AVA. Torralba and Efros [12] showed that models trained on one dataset often fail to generalize to others due to biases in scale, viewpoint, and photographic style, a problem directly applicable to aesthetic assessment. Shankar et al. [13] further demonstrated that large visual datasets are heavily skewed toward North America and Europe, causing models to perform worse on images from regions such as Africa or South Asia. De Vries et al. [14] confirmed similar geographic disparities in object detection. In aesthetic assessment specifically, Kairanbay et al. [15] found that demographic factors influence aesthetic preferences. For example, East Asian participants tended to prefer higher contrast and saturation than Western participants. However, no study has yet systematically examined how an AVA-trained model performs on non-Western photographs across multiple diagnostic dimensions, which is the gap this paper addresses.

C. Visual Attention and Explainability

Explaining why a model assigns a particular aesthetic score remains an open challenge [11]. Grad-CAM [4] offers a practical solution by generating spatial heatmaps that highlight which image regions most influenced the model’s prediction. Unlike the earlier CAM method [16], which required specific architectural changes, Grad-CAM works with any CNN architecture. In aesthetic assessment, such attention maps have been used to identify visually important regions [17], though mostly as qualitative illustrations rather than quantitative tools. Entropy-based metrics can measure how focused or spread out a model’s attention is useful for detecting domain shift, where uncertain models may attend more diffusely. This paper uses both border-to-centre ratio and attention entropy as quantitative diagnostics to compare model behaviour on Western versus non-Western images.

III. DATASETS

A. Training Data: AVA

The model is trained on the Aesthetic Visual Analysis (AVA) dataset [1], a large collection of photographs from an online photography community. Each image comes with a distribution of human aesthetic ratings on a scale from 1 to 10.

A reproducible subset of 23,198 images was sampled from AVA for fine-tuning. The sampling is stratified across ten equal-width bins of mean aesthetic score under a fixed random seed. Stratification matters because AVA’s mean scores are heavily concentrated near the middle of the scale: the two lowest and two highest bins contain only 104, 738, 1,287, and 69 images respectively. The sampler retained every image available in those sparse tail bins and drew 3,500 from each of the well-populated middle bins. The resulting subset was split into training (18,558), validation (2,319), and test (2,321) sets in an 80/10/10 ratio. A subset is sufficient here because we fine-tune a pretrained ResNet18, which needs much less data than training from scratch, and because the goal is a diagnostic study rather than the highest possible in-distribution accuracy.

B. Non-Western Evaluation Data

To assess cross-cultural generalization, we curated photographic evaluation sets from Wikimedia Commons across three regional groupings: East Asia, South Asia, and the Middle East and North Africa (MENA). To keep the medium constant with AVA and avoid mixing “cultural content” with “artistic medium,” only photographs were collected. Paintings, illustrations, and other non-photographic media were excluded.

Images were gathered by recursively traversing a fixed set of 17 seed categories (6 for East Asia, 5 for South Asia, 6 for MENA) to a depth of two subcategory levels. Two layers of filtering were applied. First, during traversal, any subcategory whose title contained non-photographic terms (such as “in art,” “map,” “logo,” “diagram,” “painting,” “sculpture,” “manuscript”) was excluded, preventing the scraper from descending into branches unlikely to contain photographs. Second, each candidate image was filtered on format (JPEG

TABLE I. CURATED NON-WESTERN PHOTOGRAPHIC EVALUATION SETS.

Region	Seed categories	Final images
East Asia	6	186
South Asia	5	289
MENA	6	212
Total	17	687

and PNG only), resolution (minimum 600 pixels on the shorter edge), and license (only Creative Commons or public-domain images were retained, so that the resulting set could be openly redistributed). Images were downloaded at a maximum of 1024 pixels on the long edge.

After this automatic filtering, all retained images underwent manual review. The review removed images that were blurry, near-duplicates, off-topic, or otherwise unsuitable, and limited over-representation of any single subject type. Every image’s source category, originating URL, license, and curation decision were logged in a provenance record to support reproducibility.

From an initial pool of roughly 188,000 candidate file entries (39,426 for East Asia, 114,435 for South Asia, 34,148 for MENA), automatic filtering combined with a per-region cap yielded 1,500 images for manual review, of which 687 were kept. The final per-region counts are shown in Table I. The differing retention rates across regions (37% for East Asia, 58% for South Asia, 42% for MENA) reflect real differences in the subject composition of open-licensed photography for each region; we return to this point in Section VI.

IV. METHODOLOGY

A. Model and Training Setup

Our aesthetic regressor is a ResNet18 [2] pretrained on ImageNet, with its final fully-connected layer replaced by a 10-unit linear layer followed by a softmax. The model therefore predicts a probability distribution over the discrete aesthetic scores 1 to 10, rather than a single scalar score, following the NIMA formulation [3].

The model is fine-tuned with the Earth Mover’s Distance (EMD) loss between predicted and ground-truth score distributions. EMD respects the fact that aesthetic scores are ordered (a 7 is closer to an 8 than to a 1), and is the standard loss for NIMA-style models. We use the Adam optimizer with two different learning rates: 10^{-4} for the pretrained backbone and 10^{-3} for the newly initialized classification head, with weight decay 10^{-5} and mini-batch size 128. Mixed-precision (fp16) training is used to speed up computation on the available NVIDIA RTX A5000 GPU. Images are resized to 224×224 ; during training they are also randomly cropped (scale 0.7 to 1.0) and horizontally flipped, while validation and test images use a fixed resize and center crop. All images are normalized with ImageNet statistics. The model is trained for 10 epochs, and the checkpoint with the lowest validation EMD loss is kept.

B. The Frozen Instrument and Diagnostic Measures

After training, the chosen checkpoint is frozen: it is not changed again. All later analyses on both AVA-test and the non-Western evaluation sets use this same fixed model. The frozen model is a measuring instrument; the contribution of this paper is what its behaviour reveals, not the model itself.

We examine the frozen model along three axes:

- 1) **Predicted-score distribution shift.** For each evaluation set, we compute the distribution of predicted mean scores and compare it to the distribution on AVA-test using the Kolmogorov-Smirnov (KS) statistic and the Wasserstein distance. Because both sets are scored by the same model, any systematic shift reflects the model’s behaviour, not a shift in ground-truth ratings.
- 2) **Calibration and prediction spread.** For each image we compute the standard deviation of the predicted score distribution, a per-image measure of the model’s uncertainty. We then compare the distribution of these spreads across AVA-test and each non-Western set.
- 3) **Spatial attention.** We apply Grad-CAM [4] to the final convolutional layer of the backbone, with the predicted mean score as the target. For each image we compute two complementary metrics: the border-to-centre ratio of activation mass, which measures *where* the model attends, and the Shannon entropy of the heatmap viewed as a probability distribution over pixels (normalised to $[0, 1]$), which measures *how decisively* the model attends, independent of location. Low entropy means sharp focus on a small region; high entropy means diffuse attention. Both metrics are compared across evaluation sets.

V. PRELIMINARY RESULTS

A. Instrument Validity on AVA

Before applying the frozen model to non-Western evaluation sets, we check that it functions as a competent aesthetic regressor on its in-distribution test set. The AVA subset of 23,198 images was split 80/10/10 into training, validation, and test partitions. Training converged quickly: validation EMD loss reached its minimum at epoch 3 (0.1002) and then drifted upward, indicating mild overfitting; the epoch-3 checkpoint is kept as the frozen instrument. Training and validation curves are shown in Figure 2

On the held-out AVA test set of 2,321 images, the frozen checkpoint achieves a Spearman rank correlation of 0.733 and a Pearson correlation of 0.731 between predicted and ground-truth mean scores, with an EMD loss of 0.102 and a mean-absolute error of 0.679 on the 1 to 10 scale. These values are within the range typically reported for NIMA-style models on AVA, and comfortably above the threshold required for the model to serve as a useful diagnostic instrument. One thing worth noting is that the standard deviation of predicted mean scores (0.99) is smaller than the standard deviation of the ground-truth means (1.27): the model has the mild central tendency that is common in EMD-trained aesthetic regressors.

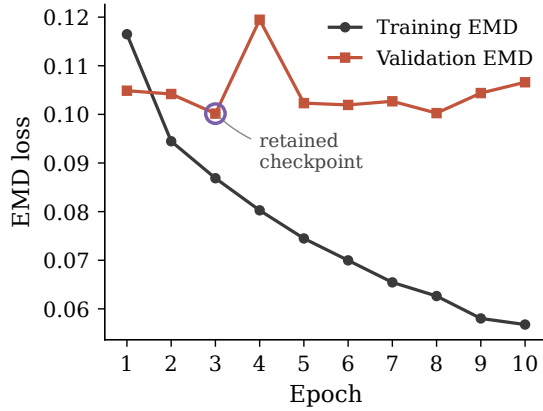


Fig. 2. Training and validation EMD loss across 10 epochs. Training loss decreases monotonically, while validation loss reaches its minimum at epoch 3 and rises mildly thereafter, indicating the onset of overfitting. The epoch-3 checkpoint is retained as the frozen instrument.

B. Diagnostic Behaviour on Non-Western Sets

1) *Predicted score distributions*: We compare the distribution of predicted mean scores on each non-Western region against the AVA-test reference using the two-sample Kolmogorov-Smirnov (KS) statistic and the Wasserstein distance. Results are summarized in Table II and shown in Figure 3. For all three regions, the predicted-score distribution differs from AVA-test at very high statistical significance ($p < 10^{-6}$ in every case), with KS effect sizes between 0.17 and 0.23. The Wasserstein distances of 0.34 to 0.41 mean that the regional distributions are on average roughly one-third to two-fifths of a score point away from AVA’s.

Two patterns stand out. First, the regional means move in different directions: East Asian photographs receive lower mean scores (5.15) than AVA-test (5.34), MENA receives higher mean scores (5.68), and South Asia is centred near AVA (5.35) but with a noticeably different distribution shape. Second, all three non-Western distributions are substantially narrower than the AVA distribution: standard deviations of predicted means are 0.59 to 0.62 across regions, against 0.99 on AVA-test. The model rarely predicts extreme scores (very high or very low) on non-Western photographs.

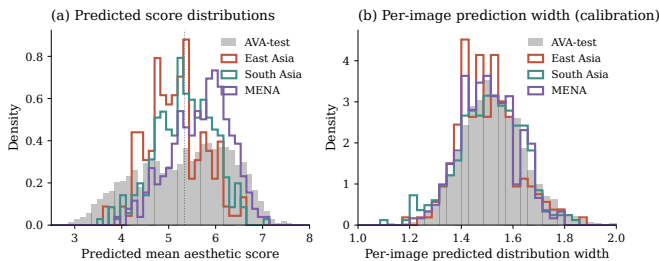


Fig. 3. Predicted mean aesthetic scores (left) and per-image predicted distribution width (right). AVA-test (grey, filled) is the in-distribution reference; the three non-Western regions (outlined) are visibly narrower and shifted in different directions. The prediction widths overlap heavily across groups.

TABLE II. DISTRIBUTION-SHIFT DIAGNOSTICS RELATIVE TO AVA-TEST. KS IS THE TWO-SAMPLE KOLMOGOROV-SMIRNOV STATISTIC; W IS THE WASSERSTEIN DISTANCE; p IS THE KS TWO-SAMPLE p -VALUE. ALL SCORE-DISTRIBUTION DIFFERENCES ARE HIGHLY SIGNIFICANT; THE CALIBRATION AXIS SHOWS A SIGNIFICANT SHIFT ONLY FOR EAST ASIA.

Region	Predicted mean score			Width	
	KS	p	W	KS	p
East Asia	0.231	1.6×10^{-8}	0.41	0.129	0.006
South Asia	0.172	4.3×10^{-7}	0.34	0.058	0.34
MENA	0.232	1.1×10^{-9}	0.40	0.081	0.15

2) *Calibration and uncertainty*: We then check whether this compression of the predicted-score range is accompanied by an increase in the model’s prediction uncertainty, measured as the standard deviation of the per-image predicted score distribution. The per-image prediction widths are essentially unchanged in absolute terms (means of 1.50 to 1.51 across regions, against 1.53 on AVA-test). Distribution-level tests reveal a mild but statistically significant calibration shift only for East Asia (KS = 0.13, $p = 0.006$); South Asia ($p = 0.34$) and MENA ($p = 0.15$) do not differ significantly from AVA. The model is therefore not visibly more uncertain on most non-Western photographs even though it is producing systematically different and more compressed score predictions on them.

3) *Spatial attention*: Grad-CAM heatmaps were computed for every image in AVA-test and the three non-Western sets, and two metrics were extracted per image as described in the methodology.

The border-to-centre ratio, after trimming the upper 5% of values per group to control for outliers, shows a statistically significant difference relative to AVA-test only for MENA (KS = 0.106, $p = 0.024$). East Asia ($p = 0.31$) and South Asia ($p = 0.09$) do not differ significantly on this metric. Where the model attends is therefore only weakly affected by the cultural domain of the input.

Attention entropy, by contrast, shows a consistent and statistically solid effect across all three regions. Median entropy is 0.907 on AVA-test and 0.911 to 0.912 on the three non-Western sets (KS = 0.116 to 0.129, p -values 0.002, 0.006, and 0.009 for South Asia, East Asia, and MENA respectively). The model’s heatmaps on non-Western photographs are systematically more diffuse than on AVA. A second observation strengthens this: while AVA entropy varies widely across images (std 0.077, spanning sharply focused to highly diffuse heatmaps), the non-Western entropies are tightly clustered (std 0.019 to 0.020). The model’s attention on non-Western photographs sits in a narrow band of moderately diffuse patterns, rather than the full range of attention behaviours it produces on AVA. These diagnostics are summarized in Table III, visualised in Figure 4, and illustrated qualitatively in Figure 5.

Qualitative inspection of the heatmaps supports a single interpretation of these numbers. On AVA images, and on the subset of non-Western photographs that have a single clearly isolated central subject, the model attends sharply to that subject. On photographs with multiple subjects, ambiguous

TABLE III. SPATIAL-ATTENTION DIAGNOSTICS RELATIVE TO AVA-TEST. BORDER-RATIO VALUES ARE COMPUTED WITH THE UPPER 5% OF EACH GROUP TRIMMED TO CONTROL FOR OUTLIERS. ATTENTION ENTROPY IS THE NORMALISED SHANNON ENTROPY OF THE GRAD-CAM HEATMAP.

Region	Border ratio (trimmed)		Attention entropy	
	KS	p	KS	p
East Asia	0.073	0.31	0.129	0.006
South Asia	0.076	0.094	0.117	0.002
MENA	0.106	0.024	0.116	0.009

focal points, or non-rule-of-thirds composition, the model defaults to attending to whatever happens to sit in the centre of the image, rather than to the meaningful photographic subject. This suggests the model has internalized a compositional prior of single-subject central framing that does not generalize uniformly across photographic traditions.

VI. DISCUSSION

A. Interpretation

The three diagnostic axes, taken together, paint a coherent picture of how an AVA-trained model behaves on non-Western photographs. The model does not simply fail or become uncertain. Instead, it produces three coordinated changes: its predicted scores shift in region-specific directions and compress across all three groups, its spatial attention becomes systematically more diffuse, and yet its per-image prediction confidence remains unchanged.

This combination means the model is, in effect, hedging its aesthetic judgments on non-Western content without signalling that it is doing so. A deployed system based on this model would produce different, compressed scores for non-Western images while appearing just as confident as it is on Western images, with no widening of the predicted distribution and no obvious cue that the model is operating on unfamiliar inputs.

We interpret this pattern as a consequence of domain shift interacting with the model’s ordinal loss function. The EMD loss encourages the model to spread probability mass

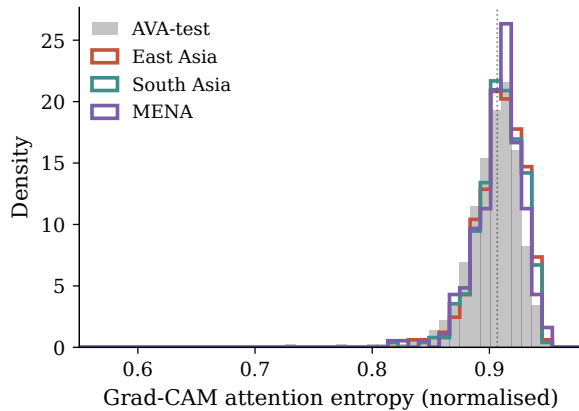


Fig. 4. Distribution of normalised Grad-CAM attention entropy per image. AVA-test (filled grey) spans the full range from focused to highly diffuse attention. All three non-Western regions occupy a narrow band of moderately diffuse attention. The dotted line marks the AVA-test median.

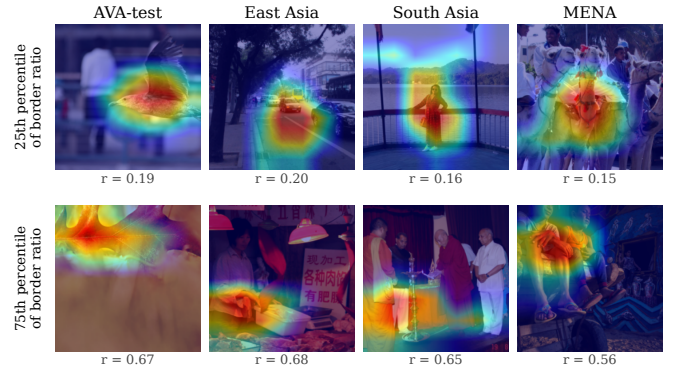


Fig. 5. Grad-CAM attention overlays for two representative images per group, selected at the 25th and 75th percentiles of the border-to-centre ratio within each group. The numerical value of the ratio is shown below each panel. AVA heatmaps range from sharply focused on a clear subject to diffuse; non-Western heatmaps tend to occupy a narrower band of moderately diffuse attention.

across nearby score levels, but it does not force the predicted distribution to become wider when the input is unfamiliar. The model has learned that safe predictions on AVA are those near the mean; when confronted with out-of-distribution inputs, it falls back to that safe region while losing the ability to localise attention sharply. The result is a model that is confidently conservative.

B. Limitations

Several limitations should be acknowledged. First, the non-Western evaluation sets have no ground-truth aesthetic ratings. We therefore diagnose behaviour, not accuracy. We cannot say whether the model’s shifted predictions are “wrong” in an absolute sense; we can only say that they differ from its behaviour on AVA. This is a deliberate design choice that trades off ground-truth availability for a larger and more culturally diverse set of images, but it remains a limitation.

Second, this study uses a single architecture (ResNet18) trained on AVA. The diagnostic framework is architecture-agnostic, and applying it to other CNN backbones (ResNet50, EfficientNet) and Vision Transformers would establish whether the compression-and-diffusion pattern is specific to ResNet18 or shared across AVA-trained aesthetic models more broadly. Similarly, repeating the diagnostic on models trained on AADB or PARA would isolate whether the observed bias originates in AVA’s specific community or is common to Western-centric aesthetics datasets in general.

Third, the regional groupings are broad and the evaluation sets modest in size (186 to 289 images per region). “East Asia” here is dominated by China, “South Asia” by India, and “MENA” by Egypt and Morocco. These groupings are not monolithic cultures, and the findings should not be interpreted as characterising aesthetic norms of entire continents. Finer-grained per-country analysis, and inclusion of currently under-represented regions such as the South Caucasus, Central Asia, and sub-Saharan Africa, would extend the diagnostic across a wider span of cultural distance from AVA. The fact that an

Azerbaijani photographic corpus of sufficient diversity could not be assembled from open-licensed sources for this study is itself a small piece of evidence for an upstream data-layer asymmetry.

Fourth, a potential confound is subject-matter composition: as noted in Section III, the regional sets differ in what is photographed. The observed mean differences may therefore partly reflect subject matter rather than a pure cultural-aesthetic bias. The paper’s claim is behavioural, and interpreting the ultimate cause requires additional, targeted data collection.

Finally, while the Grad-CAM entropy finding is statistically robust, the border-to-centre ratio shows only weak effects (significant only for MENA). The spatial attention diagnosis therefore rests primarily on the entropy metric; the border-ratio analysis is included for completeness but is not a pillar of the main claim.

C. Future Work

Three directions follow naturally from this work. First, the diagnostic framework can serve as an evaluation substrate for proposed mitigations: targeted fine-tuning on culturally balanced corpora, distribution-matching loss terms, or adversarial debiasing can be assessed by whether they push the three diagnostic axes back toward the AVA-test reference. Second, a properly designed human-rating study, drawing raters from both Western and the relevant regional communities, would let future work address whether the model’s regional bias agrees with or contradicts human aesthetic judgment. Finally, mechanistic analyses of the model’s penultimate-layer feature geometry would help link the observed behavioural compression to internal representational properties of the model.

VII. CONCLUSION

We presented a ground-truth-free diagnostic framework for examining cultural bias in AVA-trained aesthetic models and applied it to 687 curated non-Western photographs across three regions. The results are consistent across all three diagnostic axes: the model produces shifted and compressed score predictions, attends more diffusely, and shows no corresponding increase in prediction uncertainty. This combination is the kind of behaviour that ordinary in-distribution accuracy would not surface, and that real deployments would have difficulty detecting. The openly-released evaluation sets and provenance records offer a reproducible benchmark for future work on more inclusive aesthetic models.

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